

Modeling damage evolution across Antarctic ice shelves

Olivia B. Samolej¹, Jeremy M. Bassis²

¹University of Illinois Urbana-Champaign, ²CLASP University of Michigan



Introduction

Antarctic ice loss can be attributed to the melting of ice shelves

Collapse of glaciers in the Amundsen Sea Embayment, such as Thwaites, could increase sea levels by as much as 3 meters

Question: Can we simulate melt induced damage?

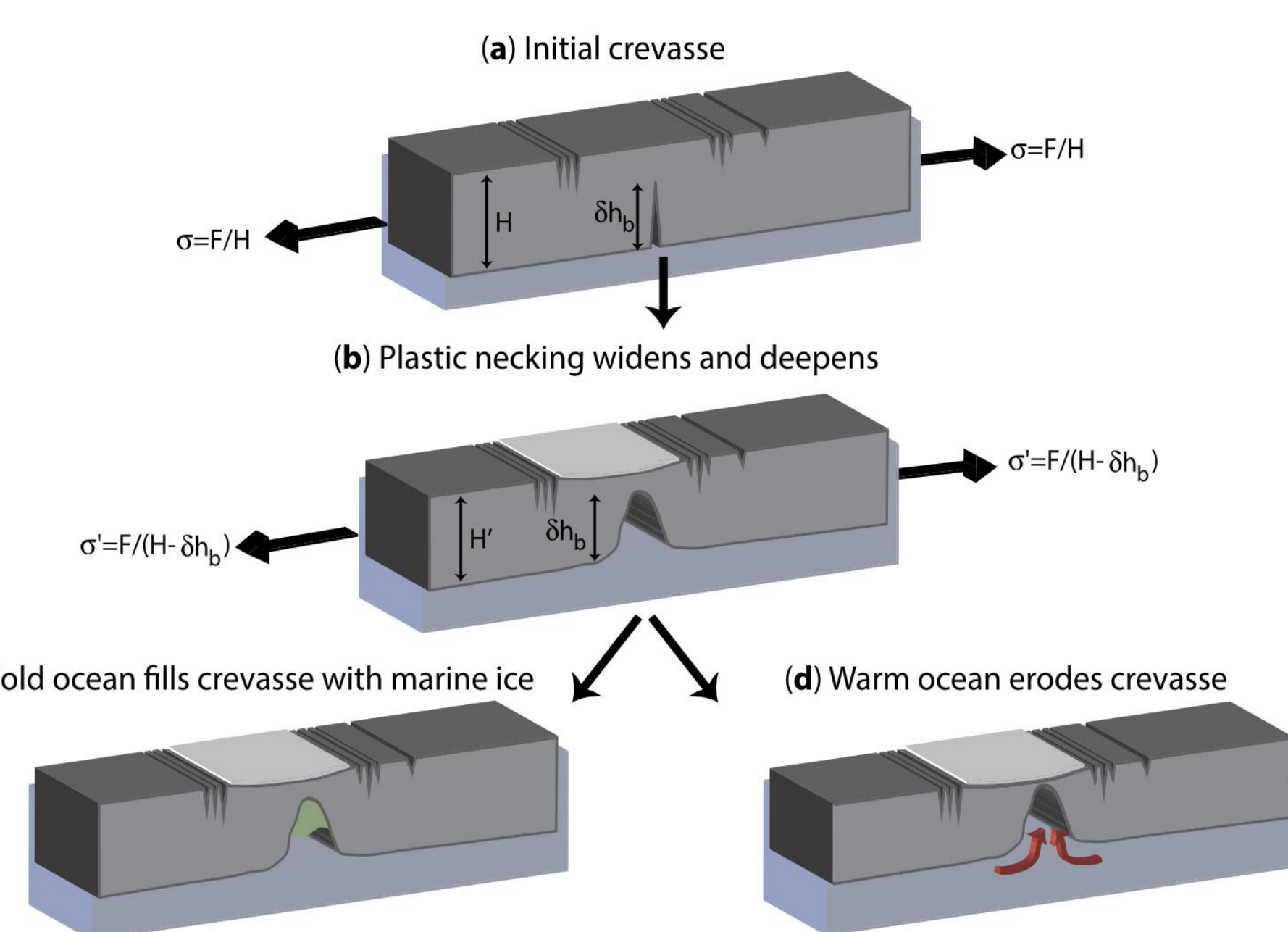


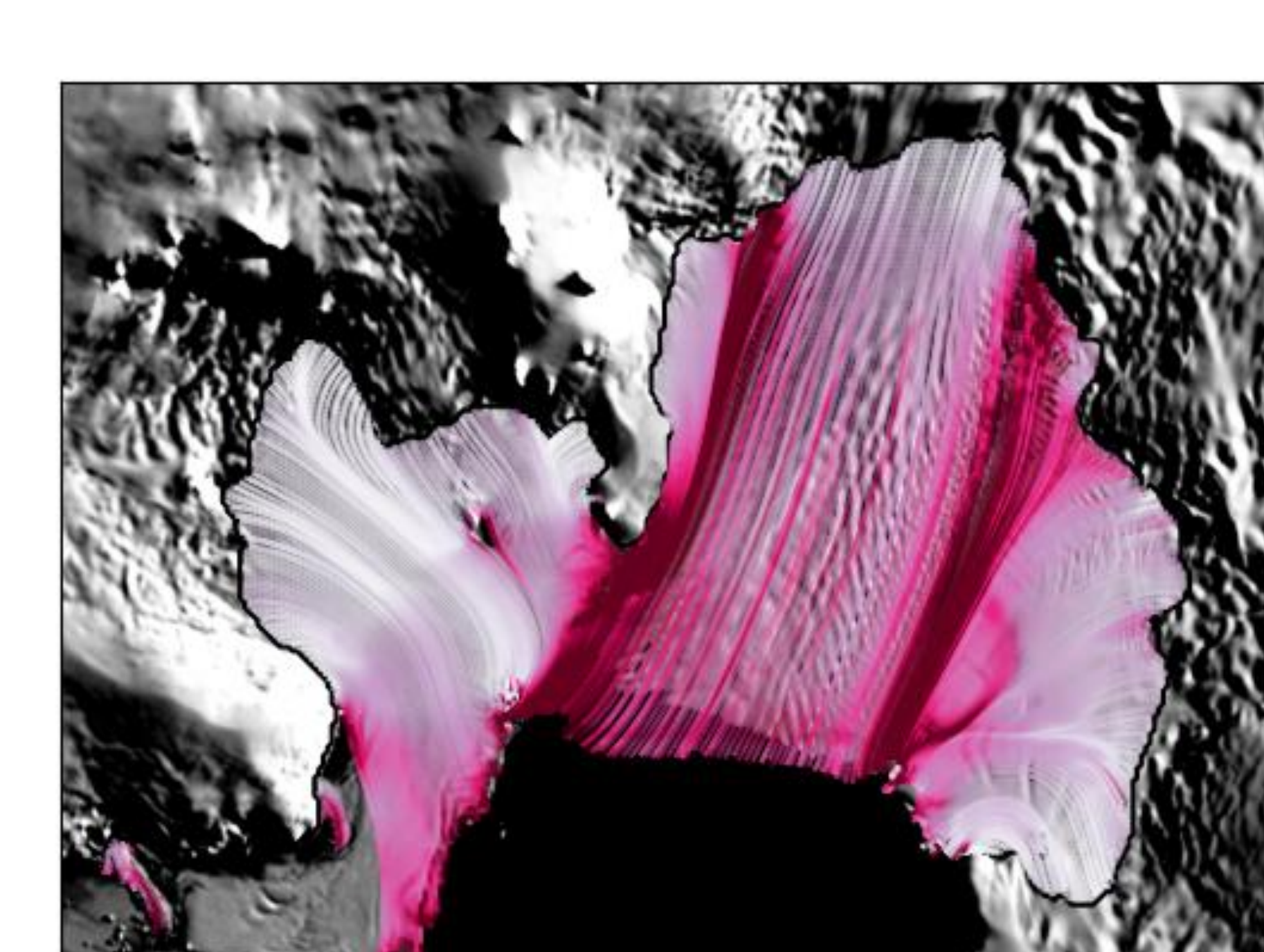
Fig 1. Schematic showing the necking instability (from Bassis et al., 2014)

- **Damage**, defined as the ratio of crevasse penetration height to total ice thickness, indicates ice shelf weakening

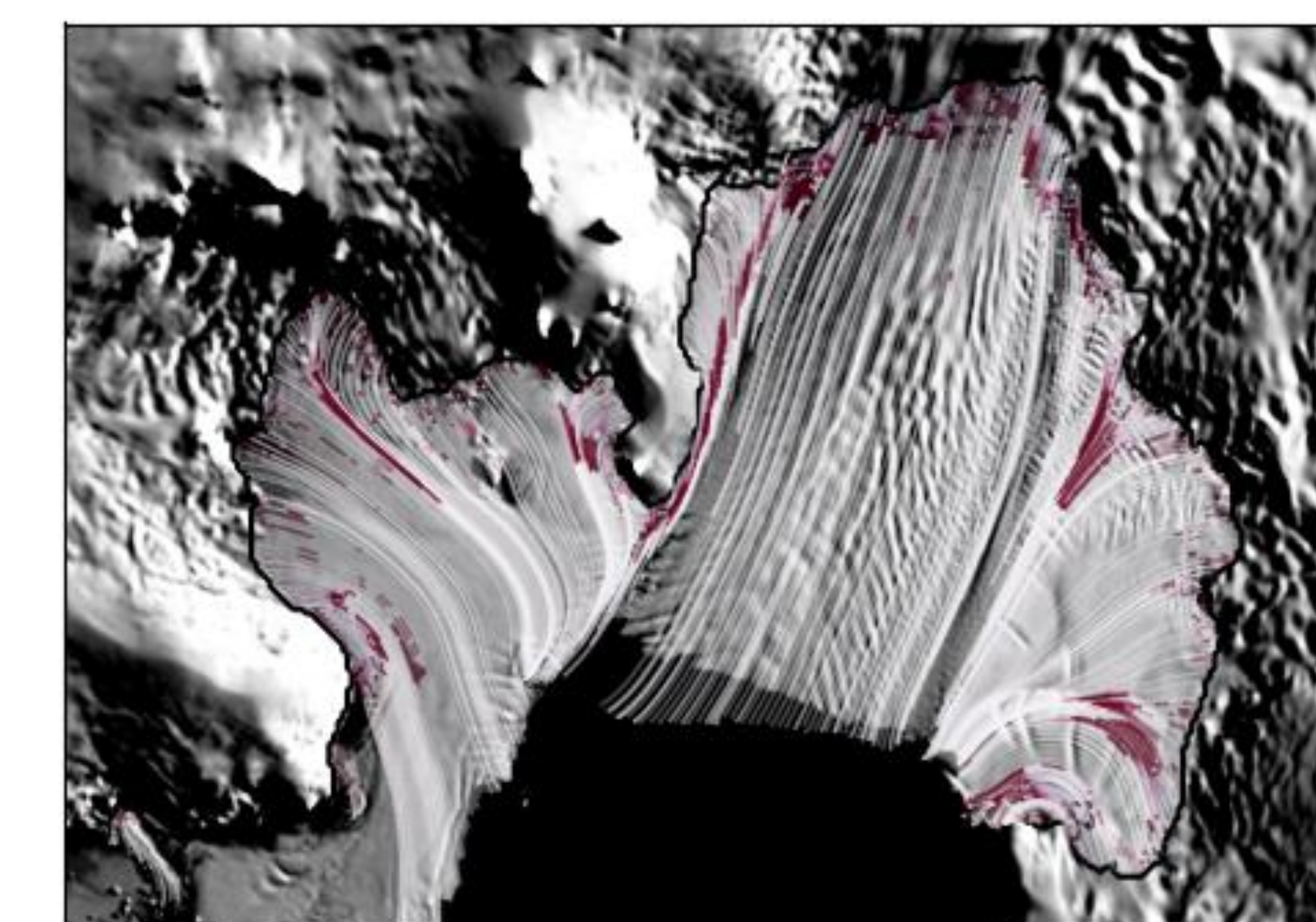
Methods

- Simulate damage evolution along flowlines using ice sheet model **BISICLES**
- We included both linear and nonlinear damage models to track how the damage evolves over time
- Compare results to roughness

Results

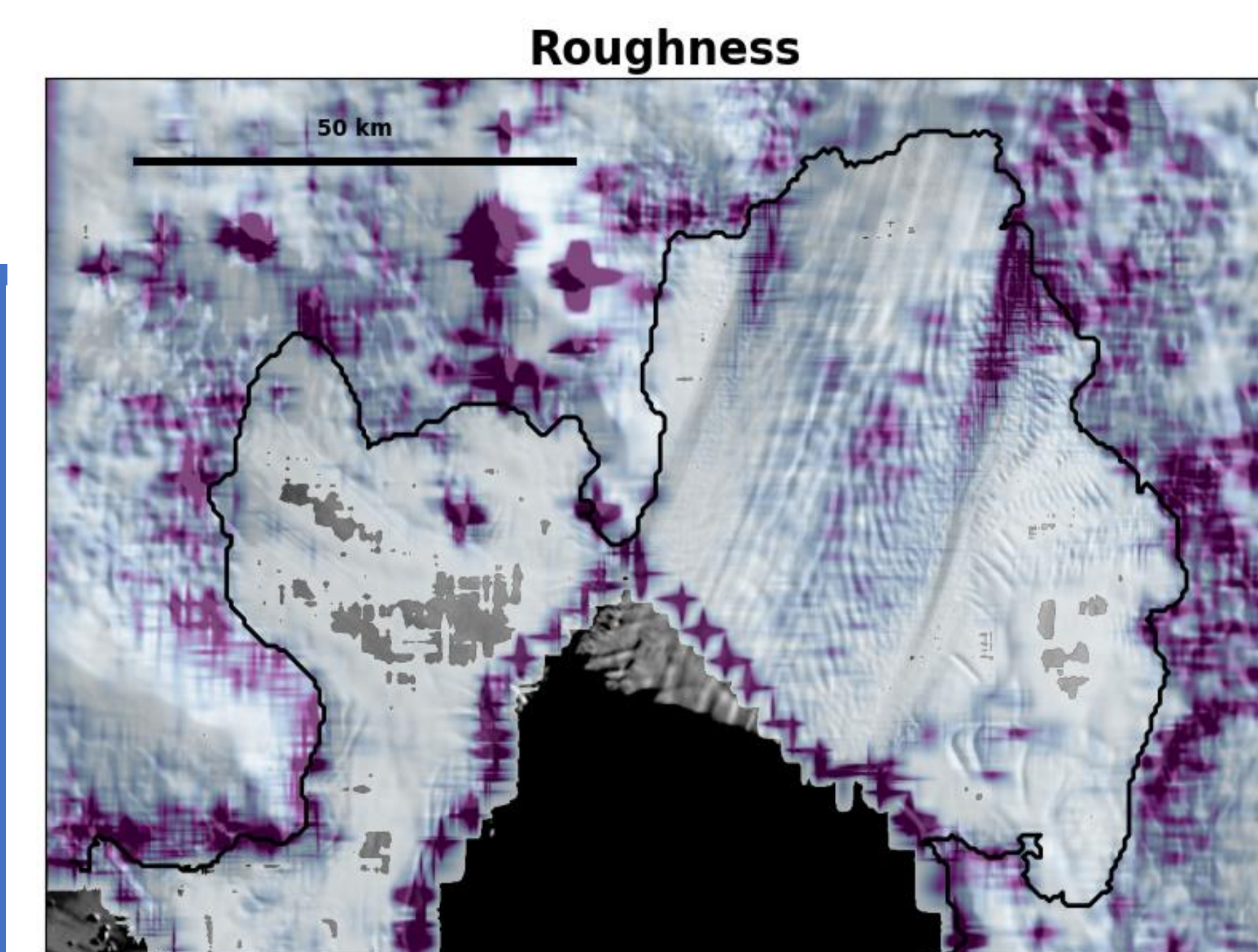


- Widespread damage

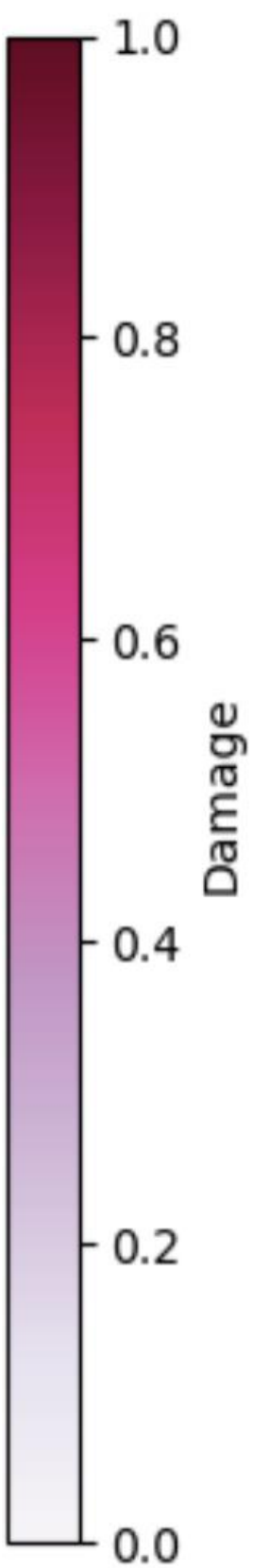
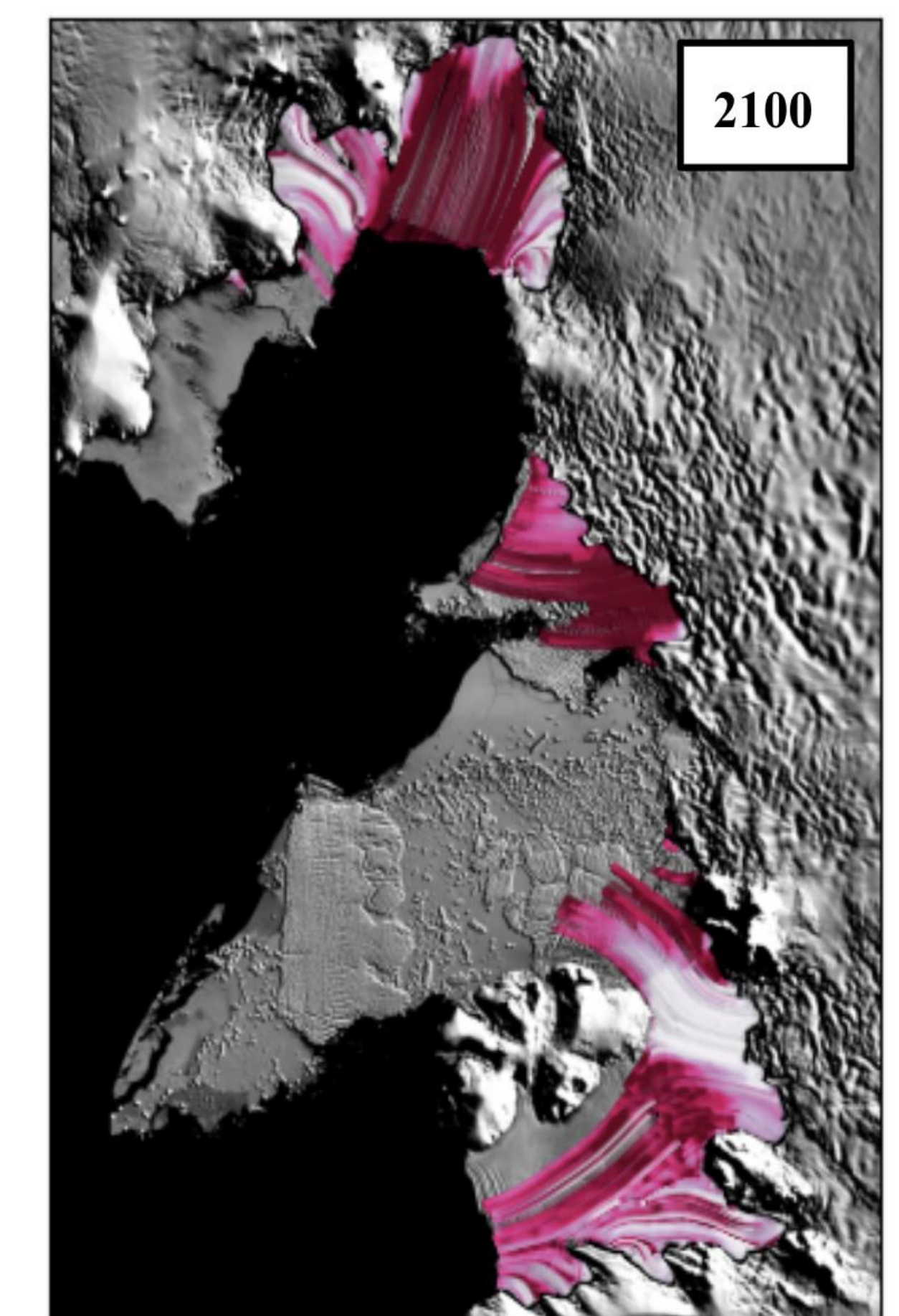
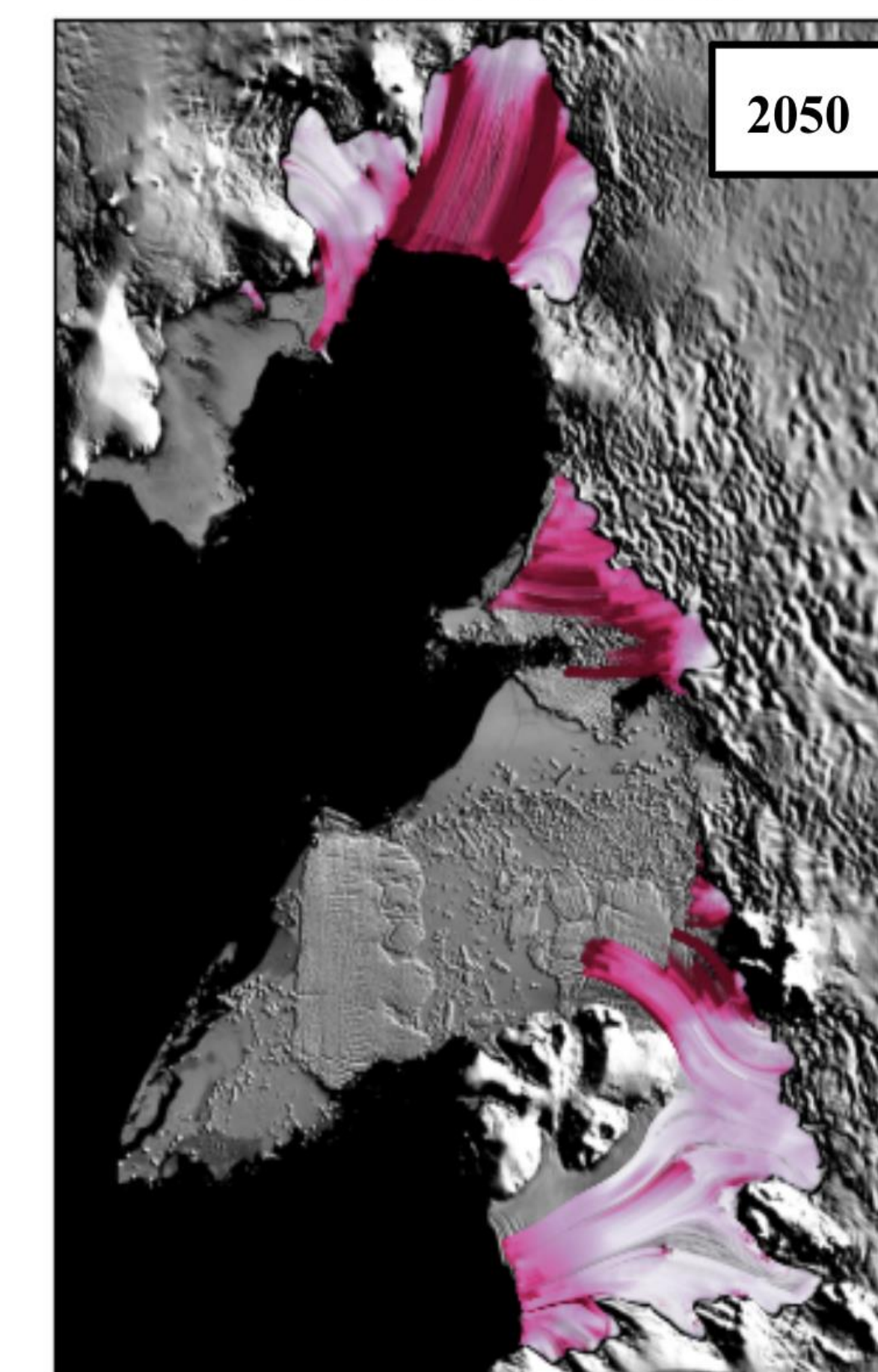
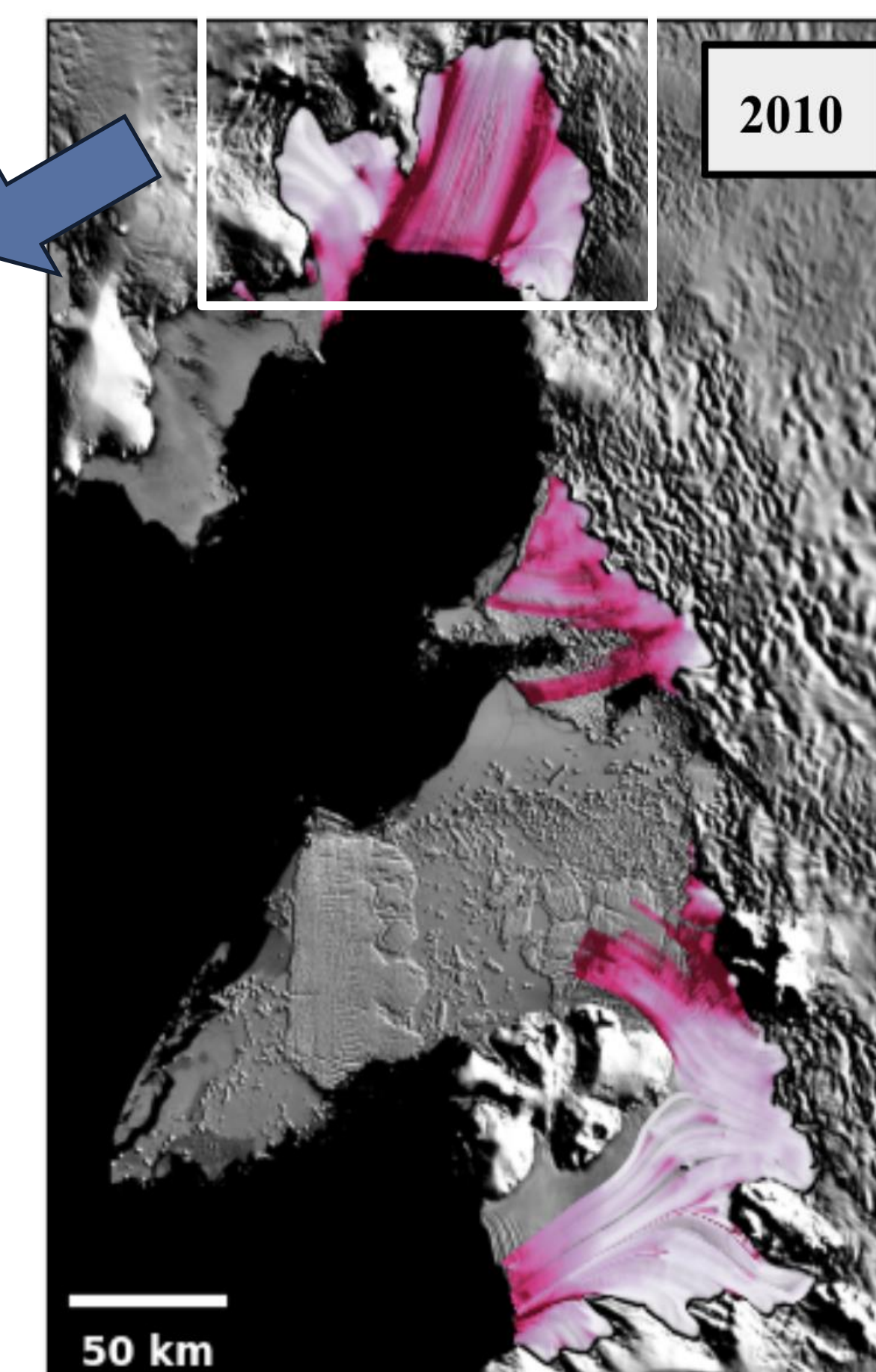


- Localized damage along shear margins

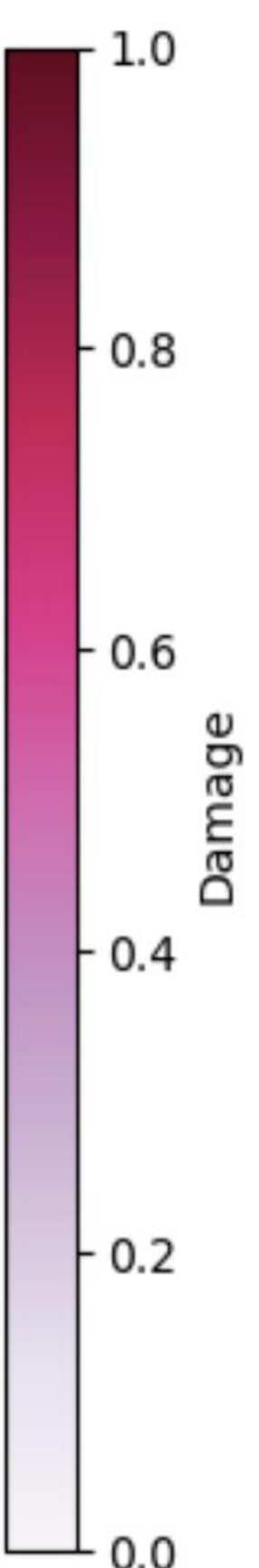
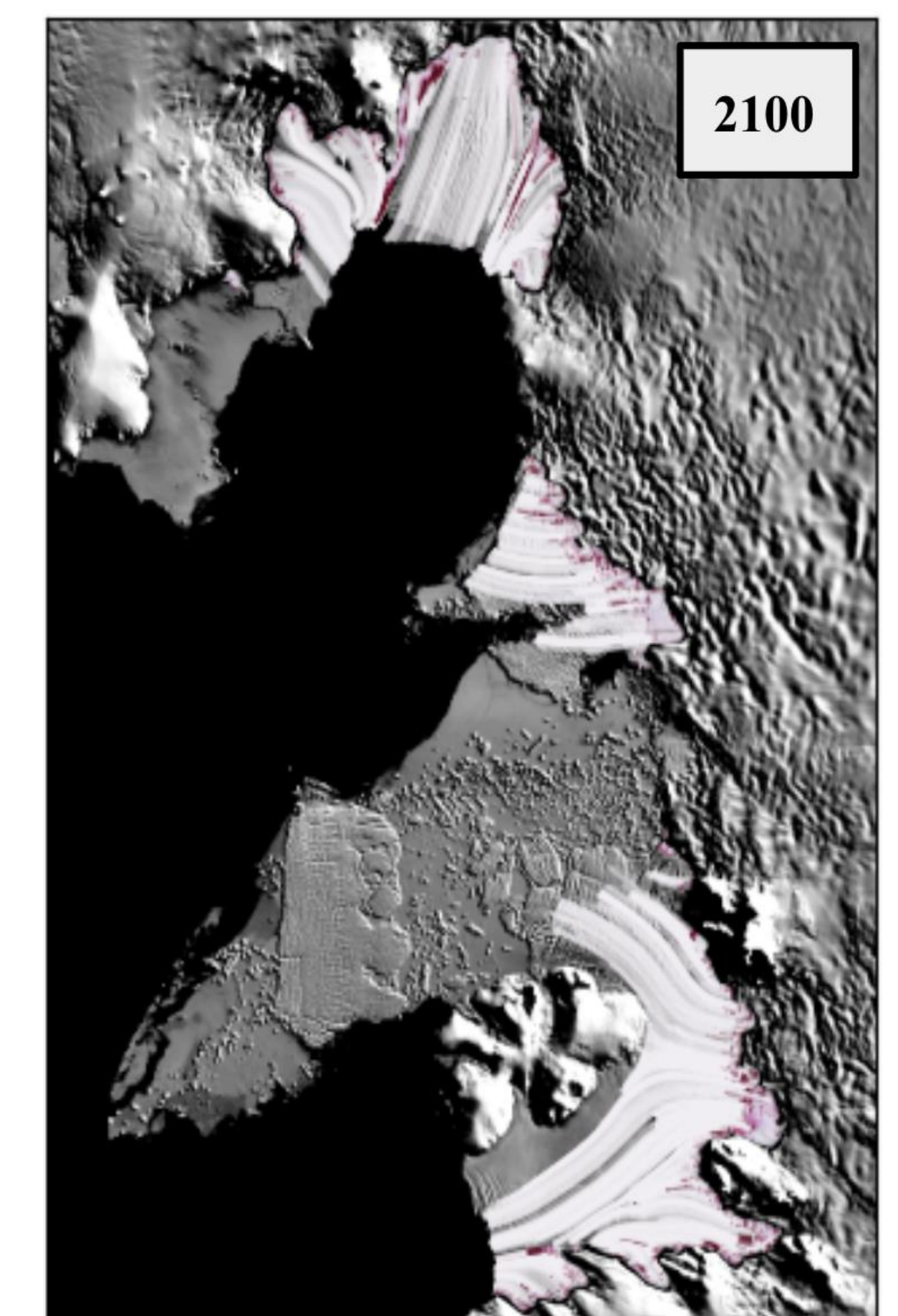
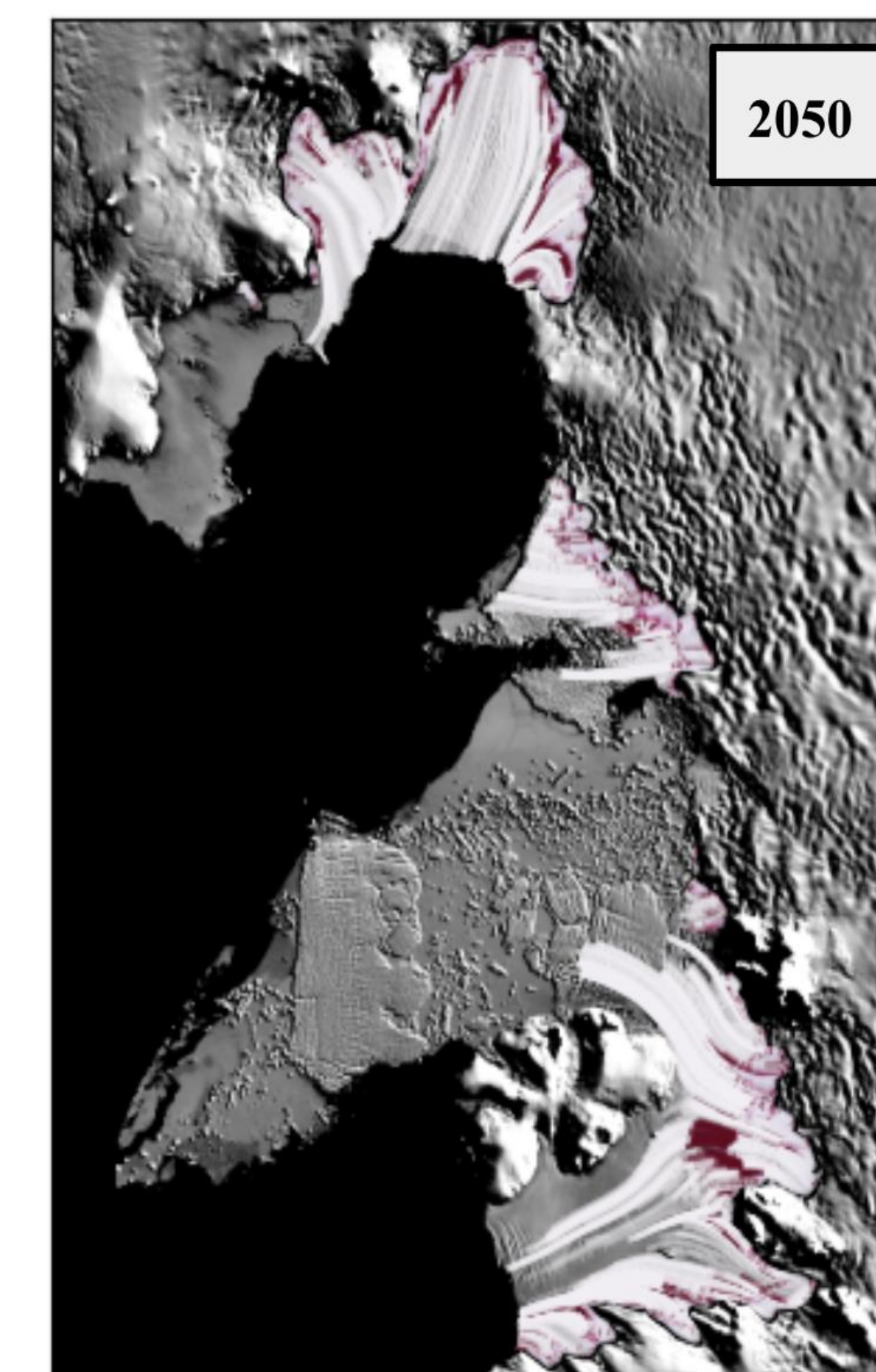
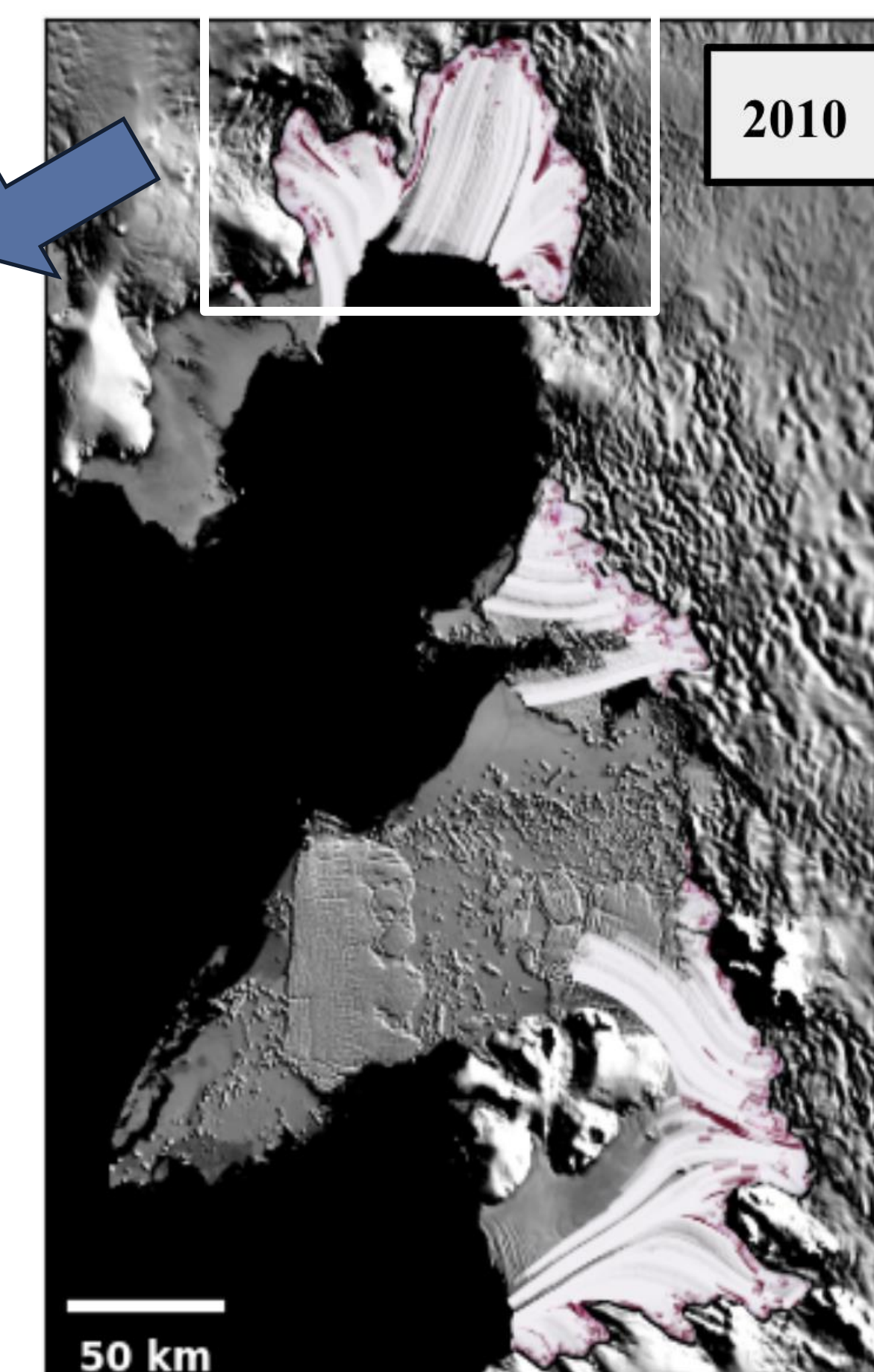
Observations



Linear Model



Non-linear Model



- **Roughness** is the variability in surface elevation (Watkins et al., 2024)
- Weakly non-linear model more closely resembles what is occurring in the surface roughness

Acknowledgements

This project was funded through a grant from the National Science Foundation's Research Experience for Undergraduates Program (Award Number 2348598).

Special thanks to Samuel Kachuck for model output and code contributions and to Ray Watkins for providing surface roughness data